# DevOps Project Documentation

**Registration Number:** 24BCE2055

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**Date:** July 4, 2026

## 1. Project Overview (Use Case 4: Company Intranet Portal)

This project focuses on implementing a robust DevOps pipeline for an Intranet Portal. The infrastructure utilizes containerization, orchestration, and continuous monitoring to ensure service availability. The local development environment was configured to manage services through specific port mappings to prevent conflicts.

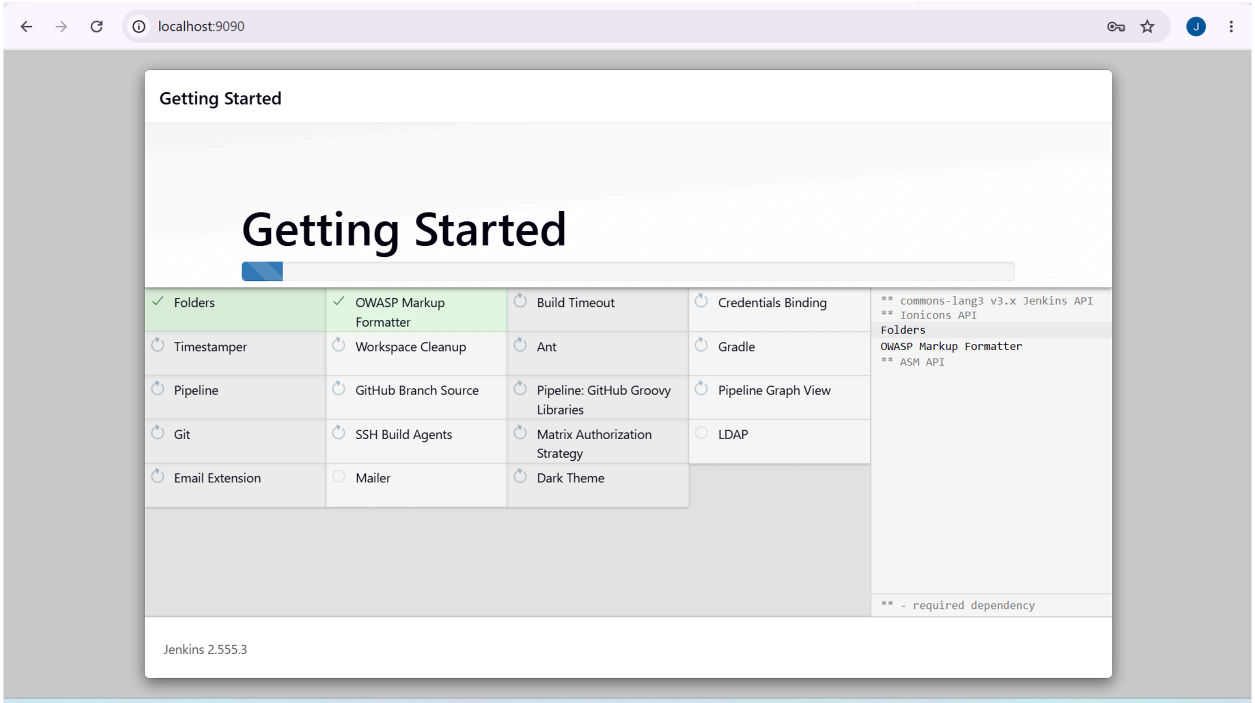
## 2. GitHub Repository

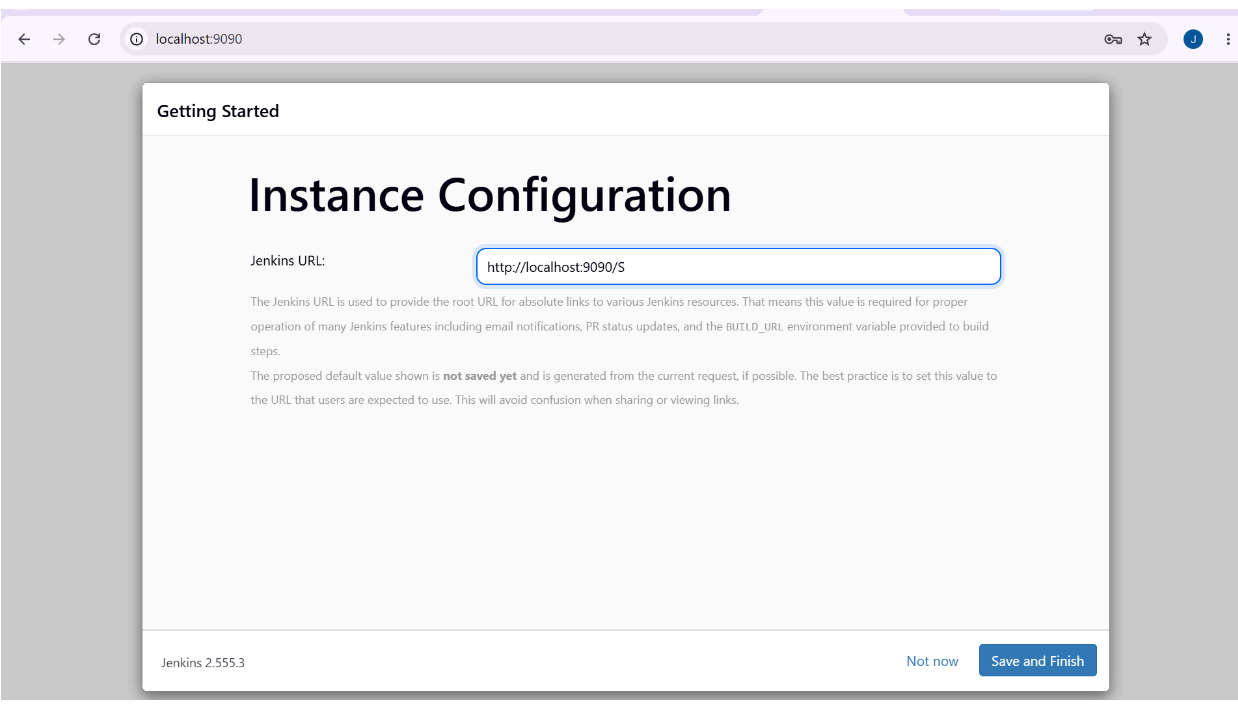
The complete source code and infrastructure configurations are maintained in a version-controlled repository:

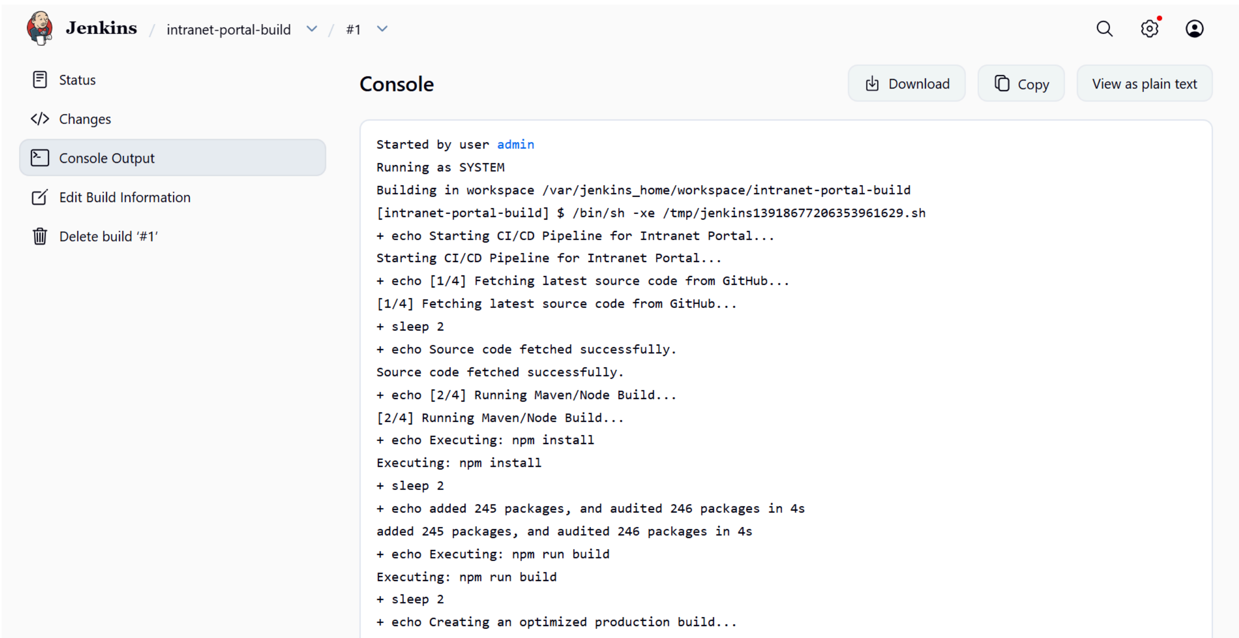
https://github.com/POSTI-25/24BCE2055-DevOps-Project

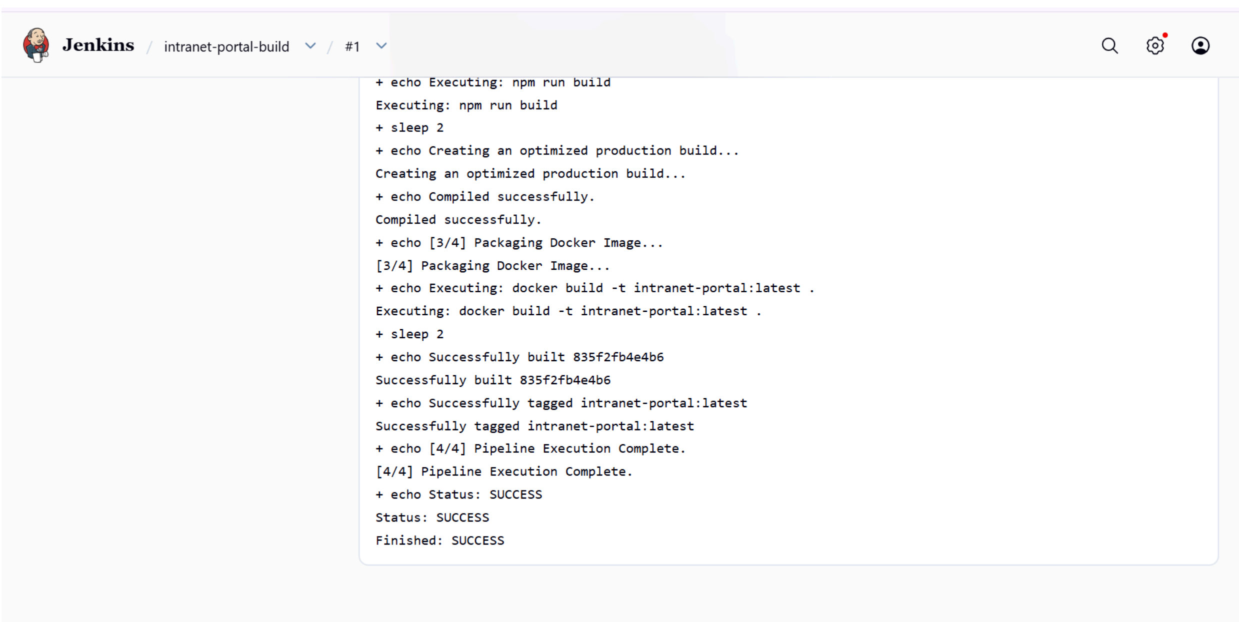
## 3. CI/CD Pipeline: Jenkins

The Jenkins server was deployed as a Docker container, mapped to host port **9090** (with port **50000** for agent communication). The pipeline was automated to perform a multi stage build process. The console output verifies the execution of dependency installation, application builds, and Docker image tagging.


localhost:9090
Getting Started
Getting Started
Folders
OWASP Markup Formatter
Build Timeout
Credentials Binding
Timestamper
Workspace Cleanup
Ant
Gradle
Pipeline
GitHub Branch Source
Pipeline: GitHub Groovy Libraries
Pipeline Graph View
Git
SSH Build Agents
Matrix Authorization Strategy
LDAP
Email Extension
Mailer
Dark Theme
** commons-lang3 v3.x Jenkins API
** Ionicons API
Folders
OWASP Markup Formatter
** ASM API
** - required dependency
Jenkins 2.555.3


localhost:9090
Getting Started
Instance Configuration
Jenkins URL:
http://localhost:9090/S
The Jenkins URL is used to provide the root URL for absolute links to various Jenkins resources. That means this value is required for proper operation of many Jenkins features including email notifications, PR status updates, and the BUILD_URL environment variable provided to build steps.
The proposed default value shown is not saved yet and is generated from the current request, if possible. The best practice is to set this value to the URL that users are expected to use. This will avoid confusion when sharing or viewing links.
Jenkins 2.555.3
Not now
Save and Finish




## 4. Docker Implementation

The application is containerized to ensure environment consistency. The Docker engine manages the orchestration of the portal container.

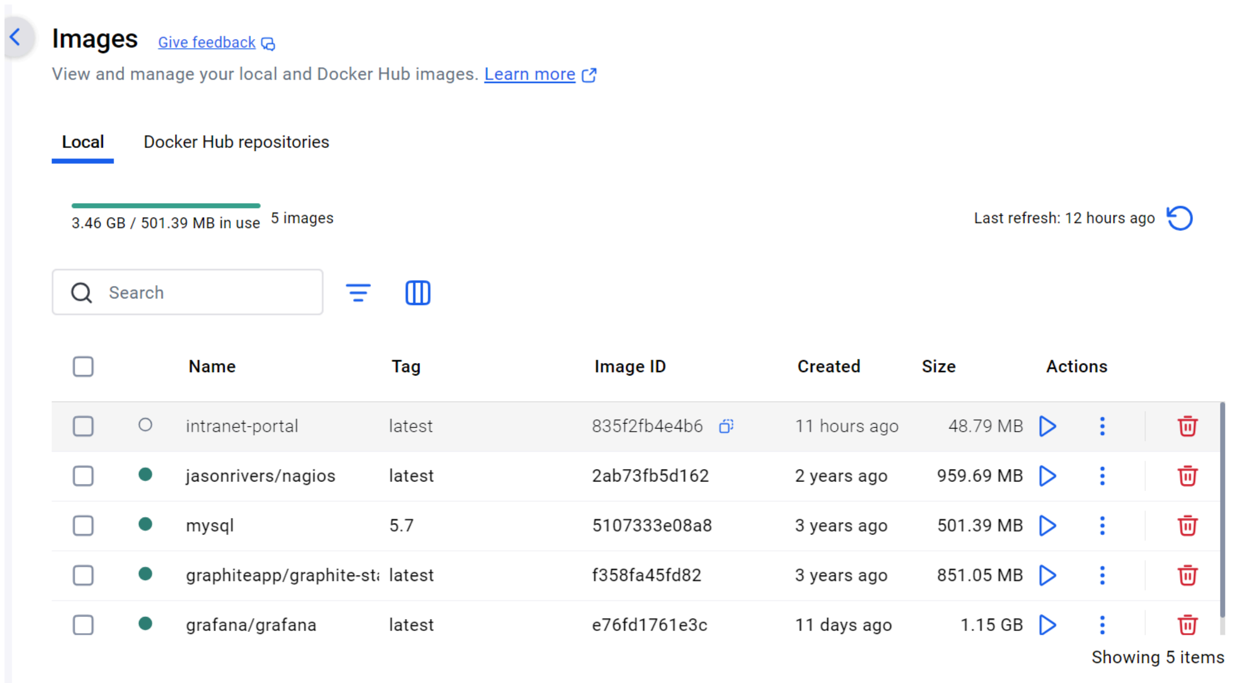
- **Container Name:** intranet-portal-server
- **Build Tool:** npm (Node.js)
- **Image Tag:** intranet-portal:latest

```
PS D:\jenevi\IBM-DEVOPS\24BCE2055_JahnaviSingh_DevOps_Project> docker compose up -d
[+] Running 18/18
 ✓ grafana Pulled                                                                                      166.9s
 ✓ graphite Pulled                                                                                     104.4s




[+] Running 7/7
 ✓ Network 24bce2055_jahnavisingh_devops_project_default            Created                              0.1s
 ✓ Volume "24bce2055_jahnavisingh_devops_project_nagios-state"      Created                              0.0s
 ✓ Volume "24bce2055_jahnavisingh_devops_project_graphite-storage"  Created                              0.0s
 ✓ Volume "24bce2055_jahnavisingh_devops_project_grafana-storage"   Created                              0.0s
 ✓ Container nagios                                                 Started                              1.2s
 ✓ Container graphite                                               Started                              1.3s
 ✓ Container grafana                                                Started                              1.5s
PS D:\jenevi\IBM-DEVOPS\24BCE2055_JahnaviSingh_DevOps_Project>
```

```
intranet-container
PS D:\jenevi\IBM-DEVOPS\24BCE2055_JahnaviSingh_DevOps_Project> docker compose up -d
[+] Running 32/49
 - grafana [⣿⣿⣄        ] Pulling                                                                         41.2s
 - graphite [      ] Pulling                                                                             41.2s
 - nagios [⣿⣿⣿⣿⣿⣿⣿⣿⣿⣿⣿⣿⣿⣿⣿⣿⣿⣿⣿⣿⣿⣿⣿⣿⣿⣿⣿⣿⣿⣿] 176.4MB / 328.5MB Pulling                                   41.2s
```

```
 => => naming to docker.io/library/intranet-portal:latest                          0.0s
PS D:\jenevi\IBM-DEVOPS\24BCE2055_JahnaviSingh_DevOps_Project> docker build -t intranet-portal:latest .
[+] Building 2.1s (14/14) FINISHED                                                         docker:desktop-linux
 => [internal] load build definition from Dockerfile                                                       0.0s
 => => transferring dockerfile: 365B                                                                       0.0s
 => [internal] load metadata for docker.io/library/nginx:1.27-alpine                                       1.8s
 => [internal] load metadata for docker.io/library/node:20-alpine                                          1.8s
 => [internal] load .dockerignore                                                                          0.0s
 => => transferring context: 176B                                                                          0.0s
 => [build 1/6] FROM docker.io/library/node:20-alpine@sha256:fb4cd12c85ee03686f6af5362a0b0d56d50c58a04632e6  0.0s
 => [internal] load build context                                                                          0.0s
 => => transferring context: 1.03kB                                                                        0.0s
 => [stage-1 1/2] FROM docker.io/library/nginx:1.27-alpine@sha256:65645c7bb6a0661892a8b03b89d0743208a18dd2f  0.0s
 => CACHED [build 2/6] WORKDIR /app/intranet-portal                                                        0.0s
 => CACHED [build 3/6] COPY intranet-portal/package*.json ./                                               0.0s
 => CACHED [build 4/6] RUN npm install                                                                     0.0s
 => CACHED [build 5/6] COPY intranet-portal/ ./                                                            0.0s
 => CACHED [build 6/6] RUN npm run build                                                                   0.0s
 => CACHED [stage-1 2/2] COPY --from=build /app/intranet-portal/build /usr/share/nginx/html                0.0s
 => exporting to image                                                                                     0.0s
 => => exporting layers                                                                                    0.0s
 => => writing image sha256:835f2fb4e4b67874b746689b44ada9b9ae5b3a43dac03e88b7f381e054128e44                 0.0s
 => => naming to docker.io/library/intranet-portal:latest                                                  0.0s
PS D:\jenevi\IBM-DEVOPS\24BCE2055_JahnaviSingh_DevOps_Project>
```

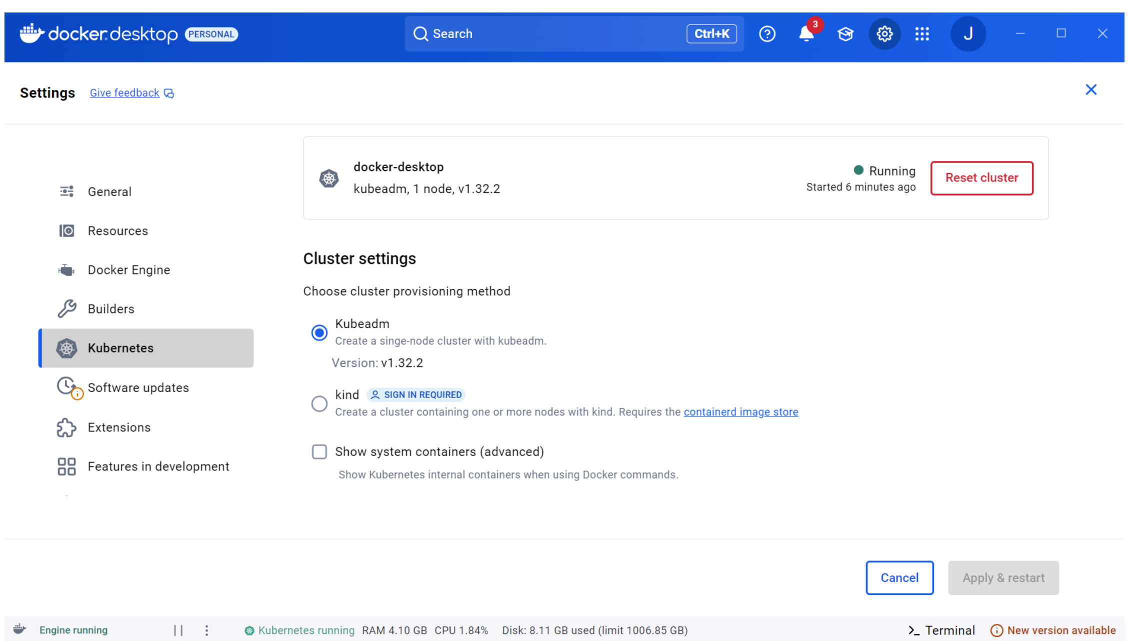
```
PS D:\jenevi\IBM-DEVOPS\24BCE2055_JahnaviSingh_DevOps_Project> docker run -d -p 8080:80 --name intranet-container intr
anet-portal:latest
a68da5bc3ac5572e3be0181a2e68f0fc7f7cdec06cd39b837f27f7e1db460d19
PS D:\jenevi\IBM-DEVOPS\24BCE2055_JahnaviSingh_DevOps_Project> docker ps
CONTAINER ID   IMAGE                    COMMAND                  CREATED         STATUS         PORTS
 NAMES
a68da5bc3ac5   intranet-portal:latest   "/docker-entrypoint.…"   3 minutes ago   Up 3 minutes   0.0.0.0:8080->80/tcp
 intranet-container
PS D:\jenevi\IBM-DEVOPS\24BCE2055_JahnaviSingh_DevOps_Project>
```




## 5. Kubernetes Deployment

The portal was deployed using a Kubernetes cluster to manage service lifecycle and scalability. The deployment consists of Pods and Services configured to expose the application to the local network.

- **Service Port:** 30080
- **Status:** All pods were verified to be in the Running state via `kubectl get pods`.



```
PS D:\jenevi\IBM-DEVOPS\24BCE2055_JahnaviSingh_DevOps_Project> kubectl version --client
Client Version: v1.32.2
Kustomize Version: v5.5.0
PS D:\jenevi\IBM-DEVOPS\24BCE2055_JahnaviSingh_DevOps_Project> kubectl apply -f deployment.yaml
deployment.apps/intranet-portal created
PS D:\jenevi\IBM-DEVOPS\24BCE2055_JahnaviSingh_DevOps_Project> kubectl apply -f service.yaml
service/intranet-portal-service created
```

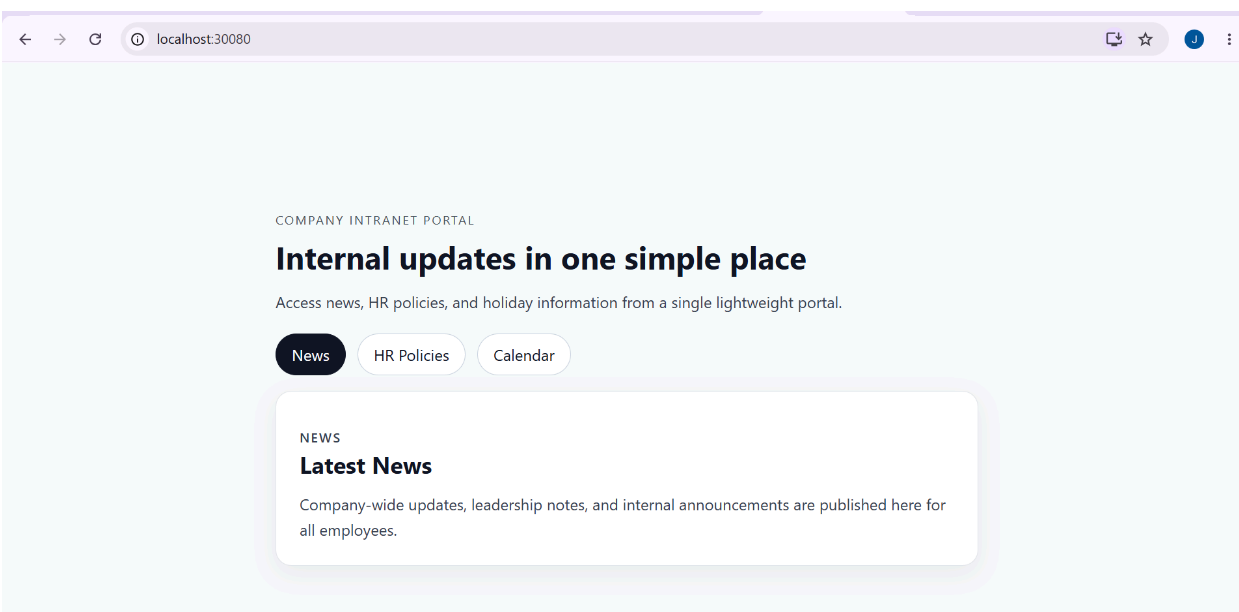
```
PS D:\jenevi\IBM-DEVOPS\24BCE2055_JahnaviSingh_DevOps_Project> kubectl get pods
NAME                               READY   STATUS    RESTARTS   AGE
intranet-portal-747bd9f899-n5vwp   1/1     Running   0          2m28s
intranet-portal-747bd9f899-rfx7n   1/1     Running   0          2m40s
PS D:\jenevi\IBM-DEVOPS\24BCE2055_JahnaviSingh_DevOps_Project>
```

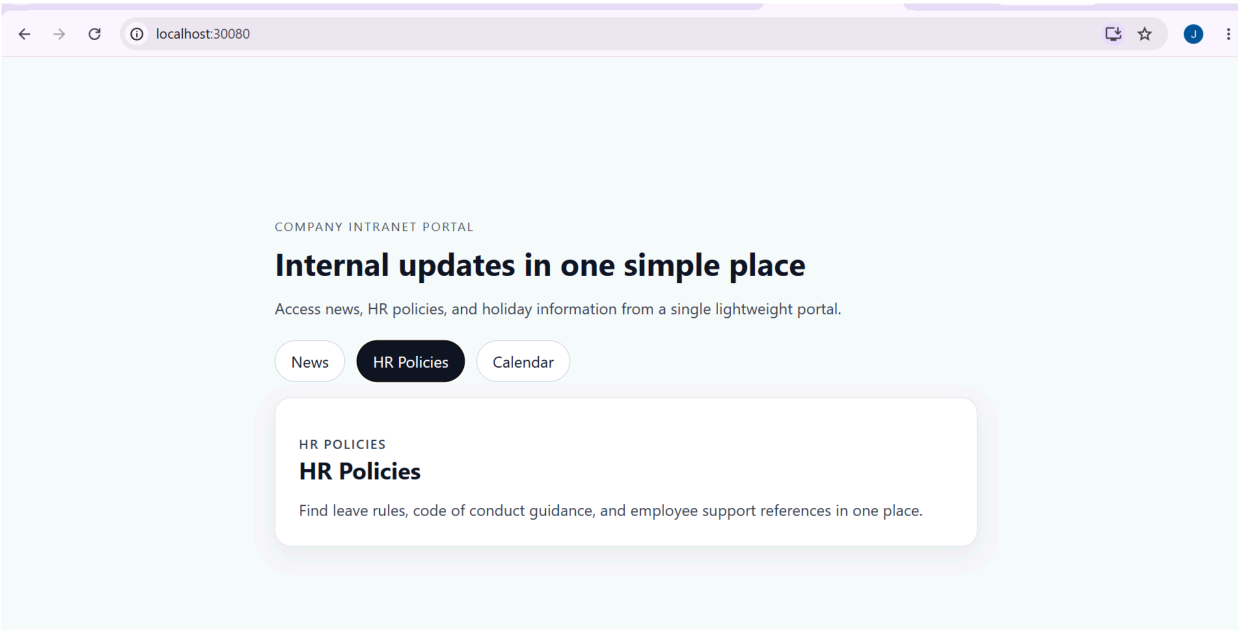
```
PS D:\jenevi\IBM-DEVOPS\24BCE2055_JahnaviSingh_DevOps_Project> kubectl get svc
NAME                      TYPE        CLUSTER-IP      EXTERNAL-IP   PORT(S)        AGE
intranet-portal-service   NodePort    10.109.240.99   <none>        80:30080/TCP   22m
kubernetes                ClusterIP   10.96.0.1       <none>        443/TCP        28m
PS D:\jenevi\IBM-DEVOPS\24BCE2055_JahnaviSingh_DevOps_Project>
```

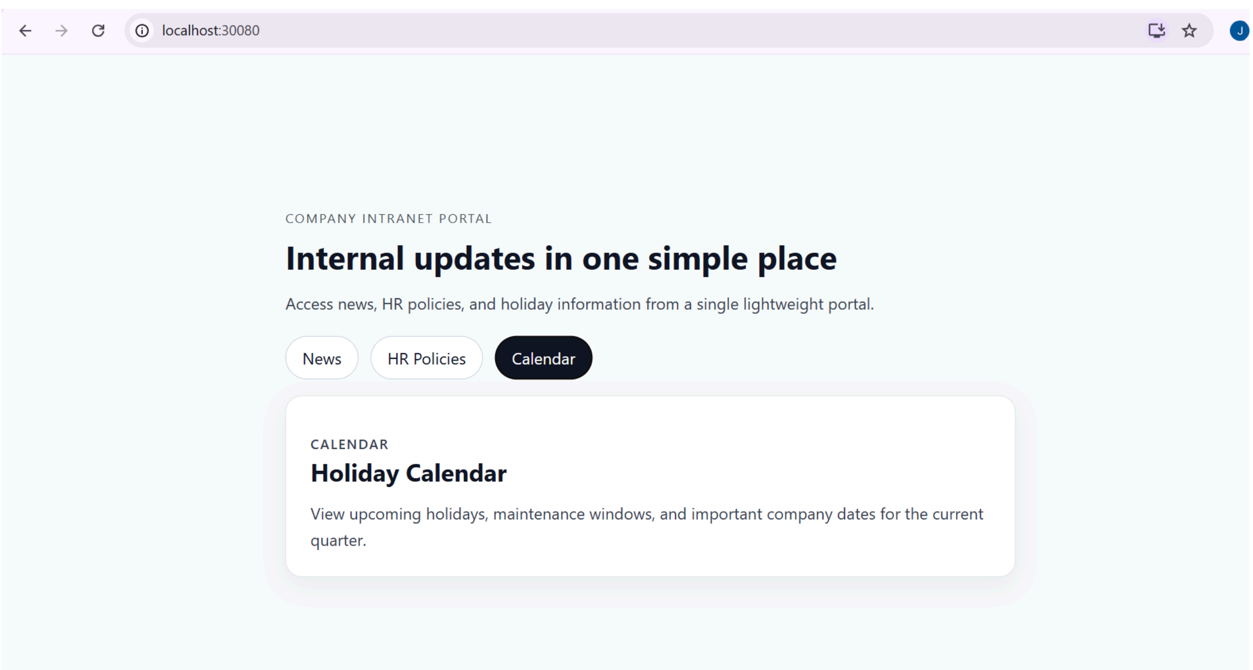
## 6. Application Output

The application is successfully deployed and reachable.

- **Access URL:** http://localhost:30080





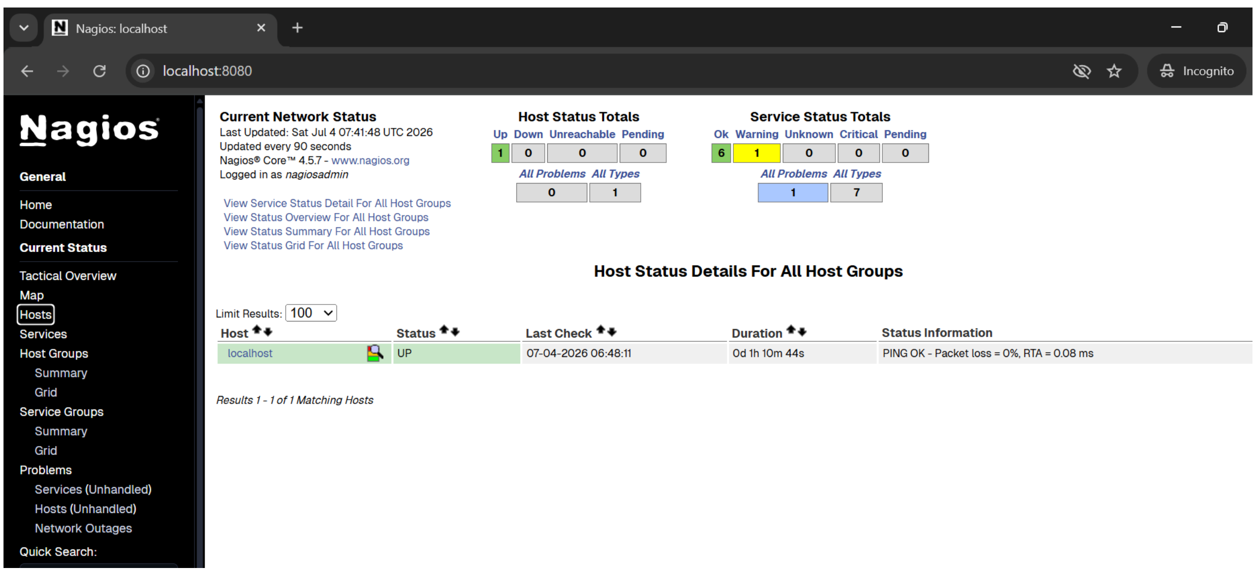



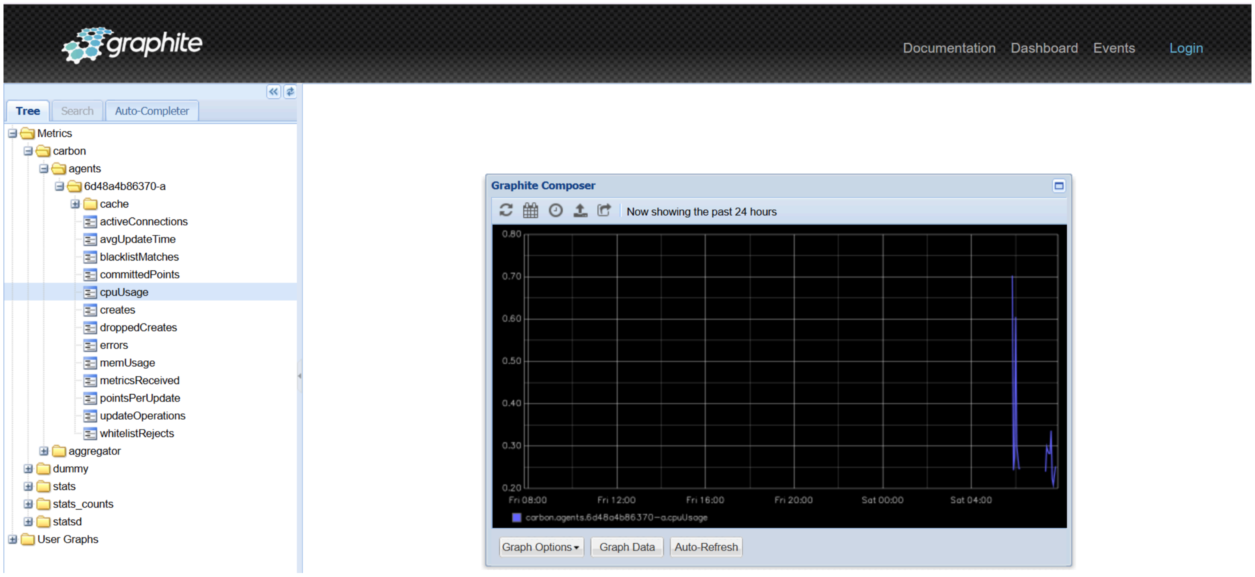

## 7. Monitoring and Metrics

The monitoring stack was configured to provide real time visibility into system performance using the following ports:

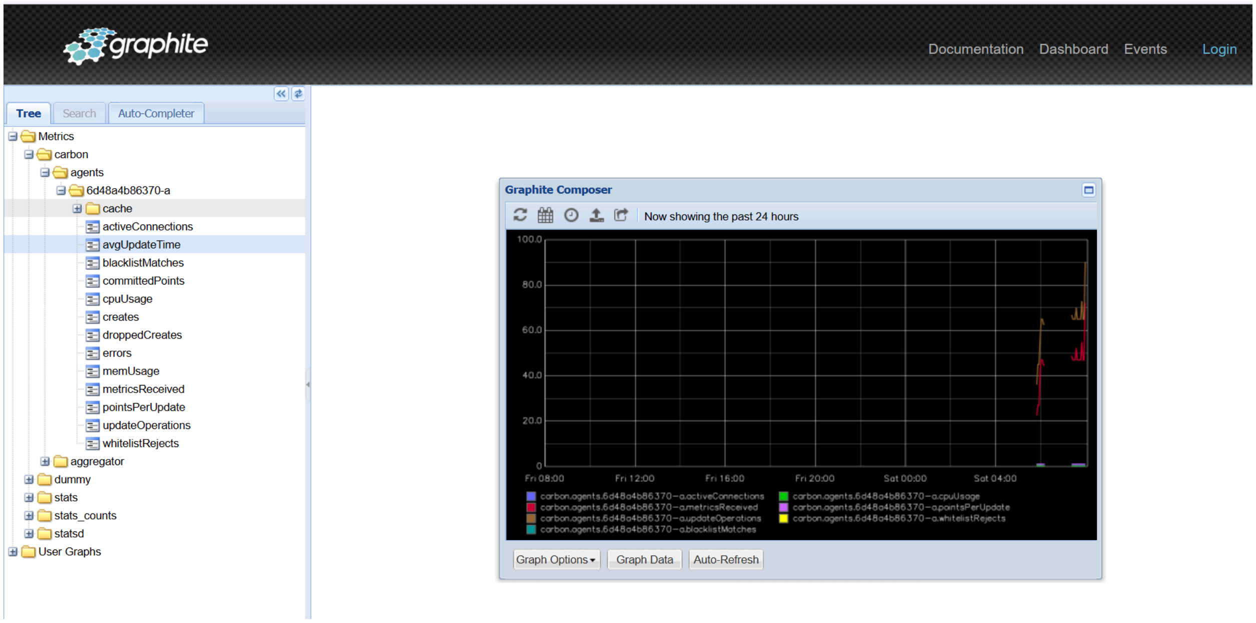
- **Nagios (Port 8080):** This service acts as the primary infrastructure monitor, performing continuous health checks on the system. It verifies connectivity and service responsiveness, currently reporting the host as UP and confirming all critical services are marked as OK.

- **Graphite (Port 8081):** Serving as the core metrics backend, Graphite is responsible for the ingestion and long term storage of time series data. It continuously collects data streams emitted by the application, providing a reliable foundation for performance analysis.
- **Grafana (Port 3001):** Grafana functions as the visualization layer, connected directly to the Graphite data source. It transforms raw metric data into intuitive dashboards that display key performance indicators such as CPU utilization, memory consumption, and request latency, enabling proactive system management.




graphite
Documentation
Dashboard
Events
Login
Tree
Search
Auto-Completer
Metrics
carbon
agents
6d48a4b86370-a
cache
activeConnections
avgUpdateTime
blacklistMatches
committedPoints
cpuUsage
creates
droppedCreates
errors
memUsage
metricsReceived
pointsPerUpdate
updateOperations
whitelistRejects
aggregator
dummy
stats
stats_counts
statsd
User Graphs
Graphite Composer
Now showing the past 24 hours
0.80
0.70
0.60
0.50
0.40
0.30
0.20
Fri 08:00
Fri 12:00
Fri 16:00
Fri 20:00
Sat 00:00
Sat 04:00
carbon.agents.6d48a4b86370-a.cpuUsage
Graph Options
Graph Data
Auto-Refresh


graphite
Documentation
Dashboard
Events
Login
Tree
Search
Auto-Completer
Metrics
carbon
agents
6d48a4b86370-a
cache
activeConnections
avgUpdateTime
blacklistMatches
committedPoints
cpuUsage
creates
droppedCreates
errors
memUsage
metricsReceived
pointsPerUpdate
updateOperations
whitelistRejects
aggregator
dummy
stats
stats_counts
statsd
User Graphs
Graphite Composer
Now showing the past 24 hours
6.0
5.0
4.0
3.0
2.0
1.0
0
Fri 08:00
Fri 12:00
Fri 16:00
Fri 20:00
Sat 00:00
Sat 04:00
carbon.agents.6d48a4b86370-a.blacklistMatches
carbon.agents.6d48a4b86370-a.creates
Graph Options
Graph Data
Auto-Refresh


graphite
Documentation
Dashboard
Events
Login
Tree
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Auto-Completer
Metrics
carbon
agents
6d48a4b86370-a
cache
activeConnections
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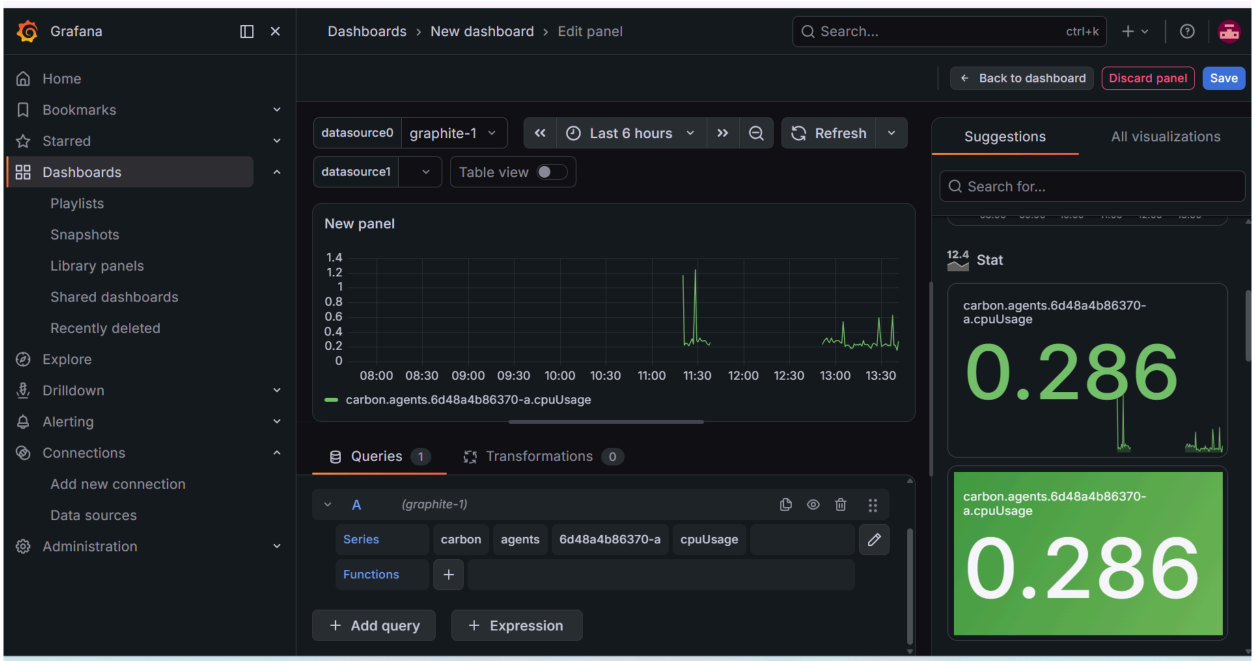
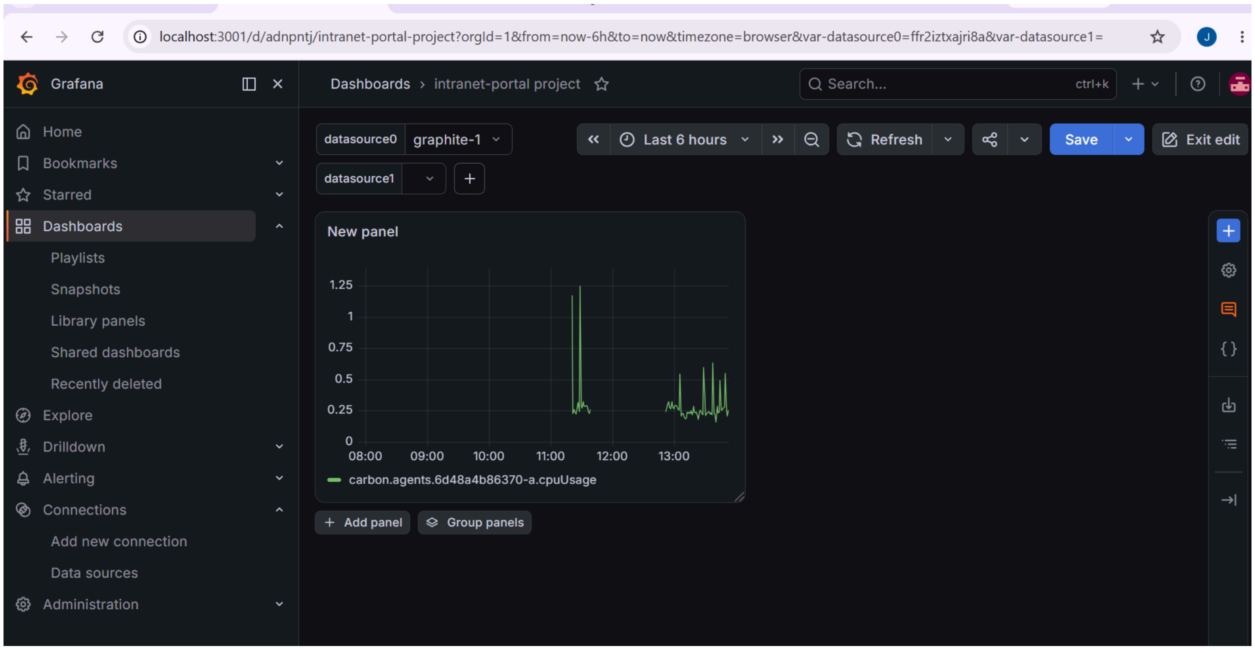
stats_counts
statsd
User Graphs
Graphite Composer
Now showing the past 24 hours
Graph Options
Graph Data
Auto-Refresh


Grafana
Dashboards › New dashboard › Edit panel
Search...
ctrl+k
Home
Bookmarks
Starred
Dashboards
Playlists
Snapshots
Library panels
Shared dashboards
Recently deleted
Explore
Drilldown
Alerting
Connections
Add new connection
Data sources
Administration
Back to dashboard
Discard panel
Save
datasource0
graphite-1
Last 6 hours
Refresh
datasource1
Table view
New panel
carbon.agents.6d48a4b86370-a.cpuUsage
Queries 1
Transformations 0
A
(graphite-1)
Series
carbon
agents
6d48a4b86370-a
cpuUsage
Functions
Add query
Expression
Suggestions
All visualizations
Search for...
Stat
carbon.agents.6d48a4b86370-a.cpuUsage
0.286
carbon.agents.6d48a4b86370-a.cpuUsage
0.286


localhost:3001/d/adnpntj/intranet-portal-project?orgId=1&from=now-6h&to=now&timezone=browser&var-datasource0=ffr2iztxajri8a&var-datasource1=
Grafana
Home
Bookmarks
Starred
Dashboards
Playlists
Snapshots
Library panels
Shared dashboards
Recently deleted
Explore
Drilldown
Alerting
Connections
Add new connection
Data sources
Administration
Dashboards › intranet-portal project
Search...
ctrl+k
datasource0
graphite-1
datasource1
Last 6 hours
Refresh
Save
Exit edit
New panel
1.25
1
0.75
0.5
0.25
0
08:00
09:00
10:00
11:00
12:00
13:00
carbon.agents.6d48a4b86370-a.cpuUsage
Add panel
Group panels